

› Taqiya Ehsan<sup>1</sup>, Luyuan Yang<sup>2</sup>, Md Musfiqur Rahman<sup>3</sup>, and Ziwei Jiang<sup>4</sup>

<sup>1</sup>Rutgers University, <sup>2</sup>University of Oklahoma, <sup>3</sup>Purdue University, <sup>4</sup>Johns Hopkins University

# SAGE: Skeptical Autonomous aGent for Experimentation



# MOTIVATION

Why should we be skeptical about agents?

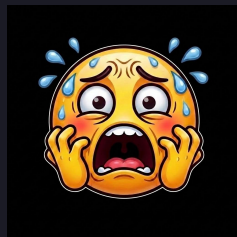
```
agent – autonomous mode
```

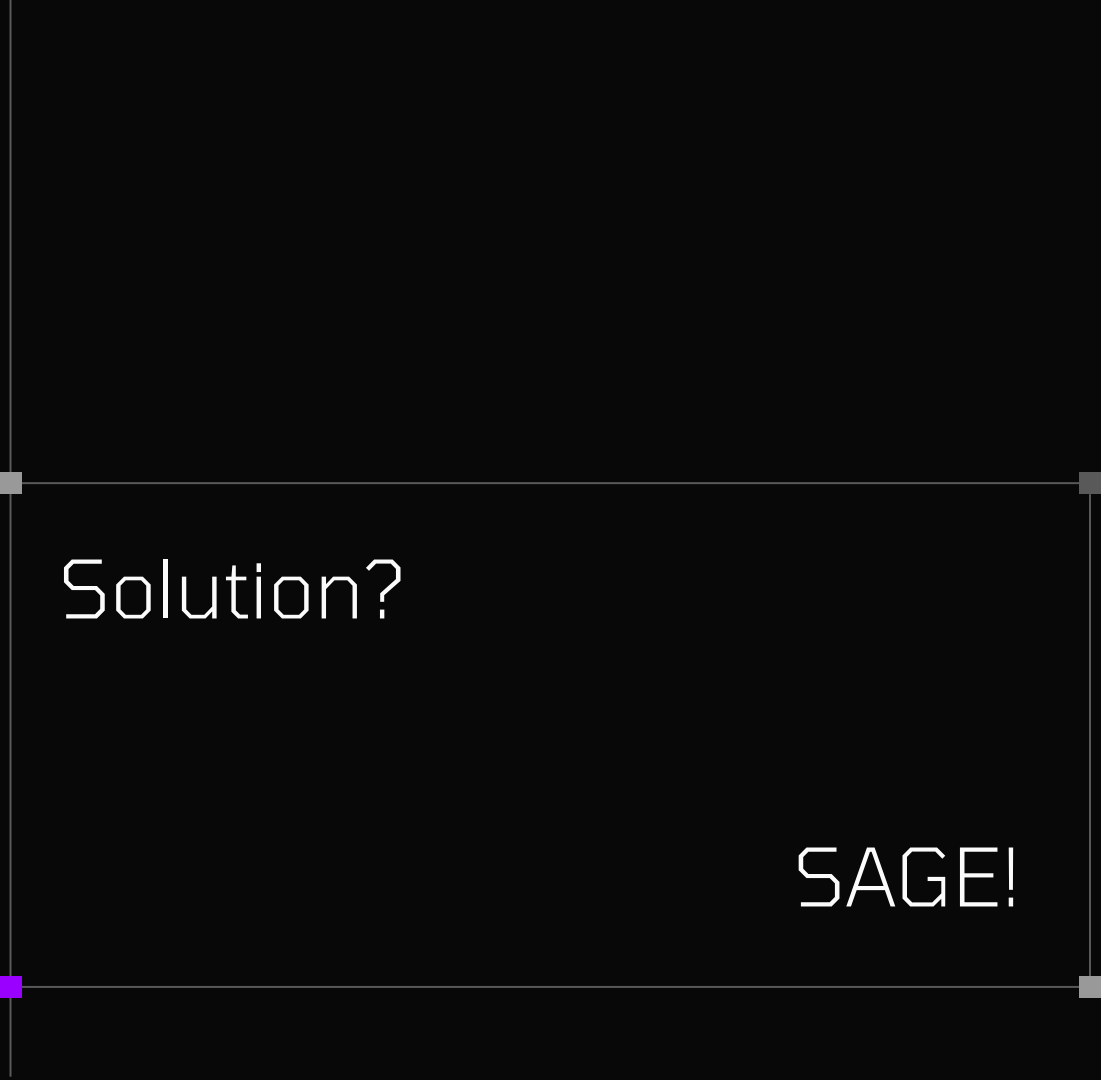
```
$ agent: cleaning up the project...
```

```
$ agent: ✓ removed 47 "unused" files
```

```
$ agent: ✓ all checks pass – done!
```

```
you: ...that was my entire src/ folder
```





Solution?

SAGE!

Demo video: [SAGE AGENT WORKFLOW](#)

# SAGE

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SKEPTICAL · AUTONOMOUS · AGENT FOR EXPERIMENTATION

A researcher agent with skeptical thinking.

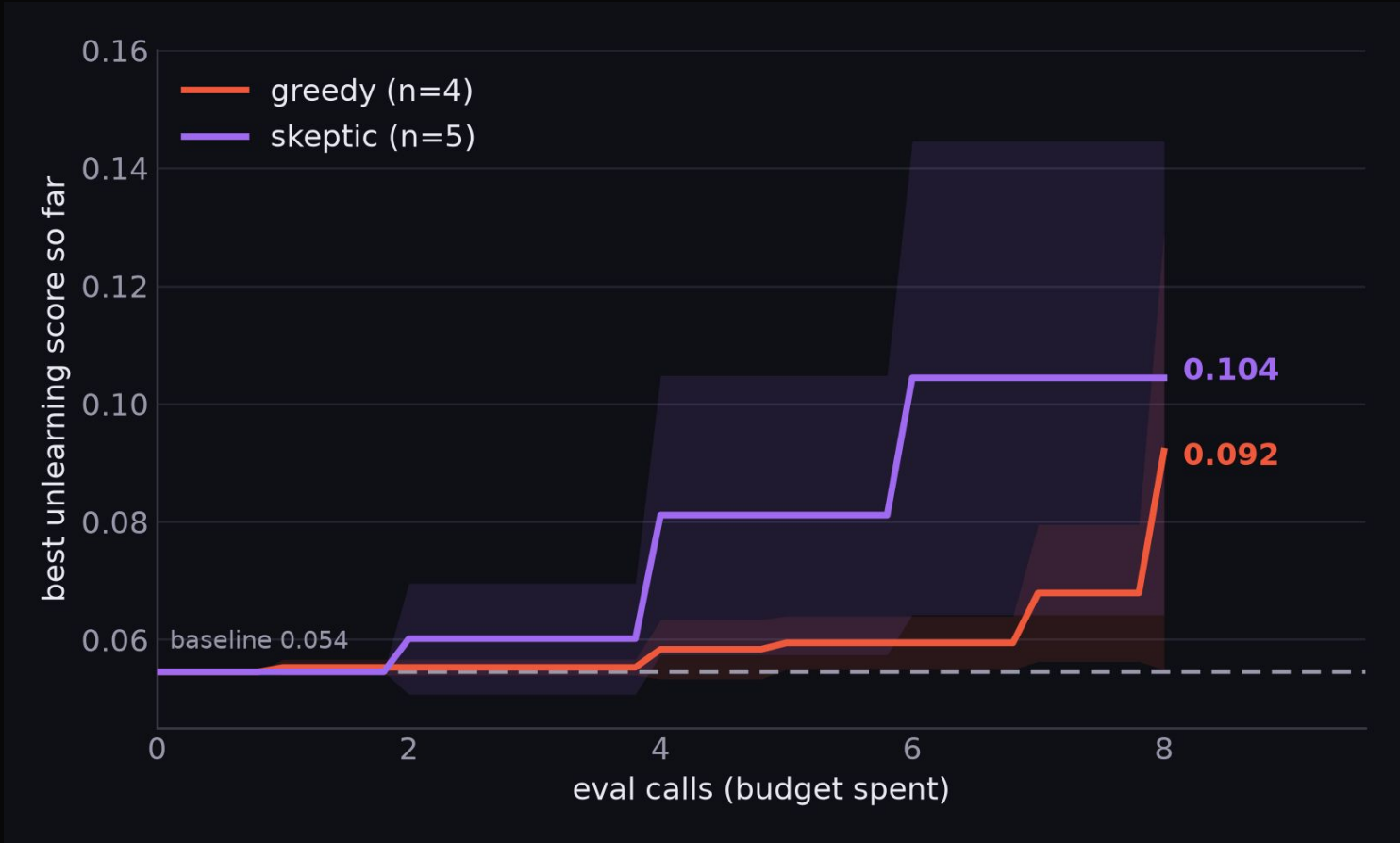
# TASKS ATTEMPTED

| Dataset                | Task                       | Domain          | Agent mode  | Result           |
|------------------------|----------------------------|-----------------|-------------|------------------|
| <b>FashionMNIST</b>    | image classification       | vision          | writes code | ✓ real gain      |
| <b>MAGIC Telescope</b> | signal vs background       | astrophysics    | writes code | ✓ real gain      |
| <b>CIFAR-10</b>        | machine unlearning         | named benchmark | writes code | ✓ skeptic wins   |
| <b>Colored MNIST</b>   | classification (spurious)  | vision          | writes code | ✗ test collapses |
| <b>Optiver</b>         | price-direction prediction | finance         | tunes HP    | ✗ no signal      |
|                        |                            |                 |             |                  |

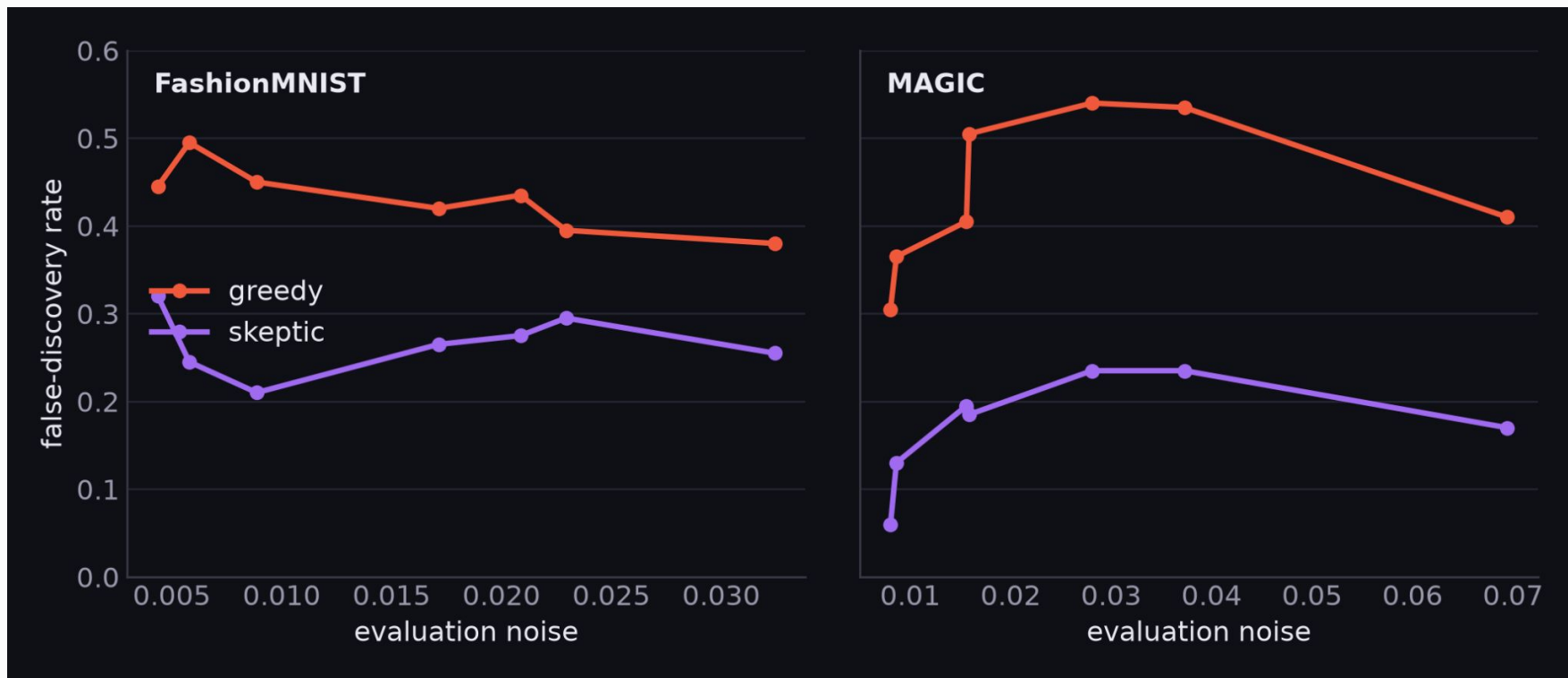
# DEVELOPMENT PROGRESS

| Task                 | Domain         | Val (base → agent) | Test (base → agent) | Outcome               |
|----------------------|----------------|--------------------|---------------------|-----------------------|
| <b>FashionMNIST</b>  | vision         | 0.74 → 0.91        | 0.75 → 0.89         | real gain             |
| <b>MAGIC</b>         | astrophysics   | 0.80 → 0.87        | 0.79 → 0.87         | real gain             |
| <b>Colored MNIST</b> | spurious-corr  | 0.61 → 0.88        | 0.58 → 0.13         | val ↑, test collapses |
| <b>Optiver</b>       | finance (tune) | 0.52 → 0.52        | 0.52 → 0.52         | no signal             |

# MLRC BENCHMARK RESULT

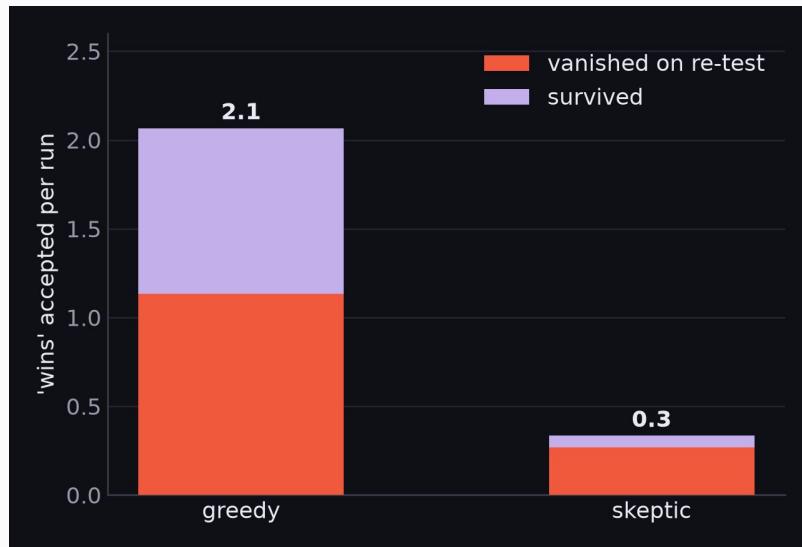
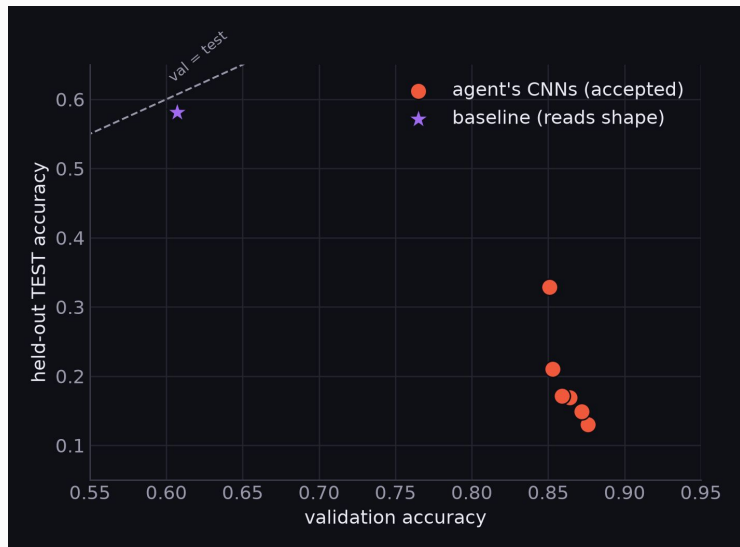


# What does the SKEPTIC do?





# FAILED EXPERIMENTS + LESSONS LEARNT



- **Colored MNIST:** the agent raises validation but *collapses on test*, a spurious cue that flips.
- **Optiver:** a near-random target, no signal to find.
- **Lessons:**
  - the skeptic re-tests over seeds, so it catches noise — **not** a win that's stable yet wrong: re-testing buys reproducibility, not necessarily validity.
  - a static check isn't enough — LLM code crashes at runtime



# TAKEAWAYS



- **Skeptic agent wins** — real progress, a better MLRC result, fooled 2–3× less under noise
- **Limitation** — can't catch distribution shift; reproducibility  $\neq$  validity
- **Next steps** —
  - shifted validation set
  - calibrated noise band
- Check out our GitHub repo → [github.com/taqiyaehsan/SAGE-2026](https://github.com/taqiyaehsan/SAGE-2026)



THANK YOU!



# References



- **MLRC-Bench** (Machine-Unlearning benchmark) — [github.com/yunx-z/MLRC-Bench](https://github.com/yunx-z/MLRC-Bench)
- **FashionMNIST** — Xiao, Rasul & Vollgraf, arXiv:1708.07747 (2017)
- **MAGIC Gamma Telescope** — Bock et al., UCI Machine Learning Repository (2007)
- **CIFAR-10** — Krizhevsky, *Learning Multiple Layers of Features from Tiny Images* (2009)
- **MNIST** — LeCun, Cortes & Burges, The MNIST Database of Handwritten Digits
- **Colored MNIST / Invariant Risk Minimization** — Arjovsky, Bottou, Gulrajani & Lopez-Paz, arXiv:1907.02893 (2019)
- **Optiver** — *Trading at the Close* — Kaggle competition (2023)